



CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342-G5 SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: ConsultEase

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Table of Contents

- 1. Introduction.....3
 - 1.1. Purpose..... 3
 - 1.2. Scope..... 3
 - 1.3. Definitions, Acronyms, and Abbreviations..... 3
- 2. Overall Description.....3
 - 2.1. System Perspective..... 3
 - 2.2. User Classes and Characteristics.....3
 - 2.3. Operating Environment..... 3
 - 2.4. Assumptions and Dependencies..... 3
- 3. System Features and Functional Requirements.....3
 - 3.1. Feature 1:.....3
 - 3.2. Feature 2:.....3
- 4. Non-Functional Requirements..... 3
- 5. System Models (Diagrams)..... 4
 - 5.1. ERD..... 4
 - 5.2. Use Case Diagram..... 4
 - 5.3. Activity Diagram.....4
 - 5.4. Class Diagram.....4
 - 5.5. Sequence Diagram.....4
- 6. Appendices.....4

1. Introduction

1.1. Purpose

ConsultEase allows students to request consultations online and track approval status, while teachers can manage and respond to requests efficiently. This document is intended for developers and stakeholders to understand system requirements and design.

1.2. Scope

ConsultEase is a web-based system for students and teachers. Students can register, log in, submit consultation requests, and view statuses. Teachers can log in to approve/reject requests. Protected pages cannot be accessed without authentication. Teachers cannot register; their accounts are pre-coded.

1.3. Definitions, Acronyms, and Abbreviations

- UI – User Interface
- API – Application Programming Interface
- CRUD – Create, Read, Update, Delete
- DB – Database

2. Overall Description

2.1. System Perspective

ConsultEase is a client-server web application. The React front-end communicates with a Spring Boot API backend, which interacts with a relational database to manage users and consultation requests.

2.2. User Classes and Characteristics

- Guest User – Can register (students only) or log in. Cannot access protected pages.
- Student (Authenticated User) – Can submit consultation requests, view dashboard, and manage profile.
- Teacher (Authenticated User) – Can approve/reject requests and view assigned requests. Cannot register.

2.3. Operating Environment

- Web browser (Chrome, Edge, Firefox)
- React JS front-end
- Spring Boot back-end
- MySQL / PostgreSQL database

2.4. Assumptions and Dependencies

- Users have internet access.
- React UI communicates with Spring Boot API over HTTPS.
- Teachers are pre-registered; students must register to use the system.

3. System Features and Functional Requirements

Describe each major feature of the system and its functional requirements.

3.1. Feature 1: User Registration

Description: Only students can create accounts. Teachers are pre-coded.

Functional Requirements:

- Students must provide a full name, email, and password.
- Passwords are hashed before storage.
- Role is automatically set to Student.

3.2. Feature 2: Login / Logout

Description: Students and teachers log in to access protected pages.

Functional Requirements:

- Enter username and password to log in.
- Validate credentials and return session/token.
- Logout invalidates the session/token.

3.3. Feature 3: Dashboard & Profile

Description: Authenticated users can view dashboard and profile.

Functional Requirements:

- Dashboard shows consultation requests (students: own; teachers: assigned).
- Profile allows updating personal information.

3.4. Feature 4: Consultation Request Submission (Transactional)

Description: Students submit consultation requests; teachers approve/reject them.

Functional Requirements:

- Form includes subject, description, and assigned teacher.
- Request stored with status Pending.
- Teachers can approve or reject requests.

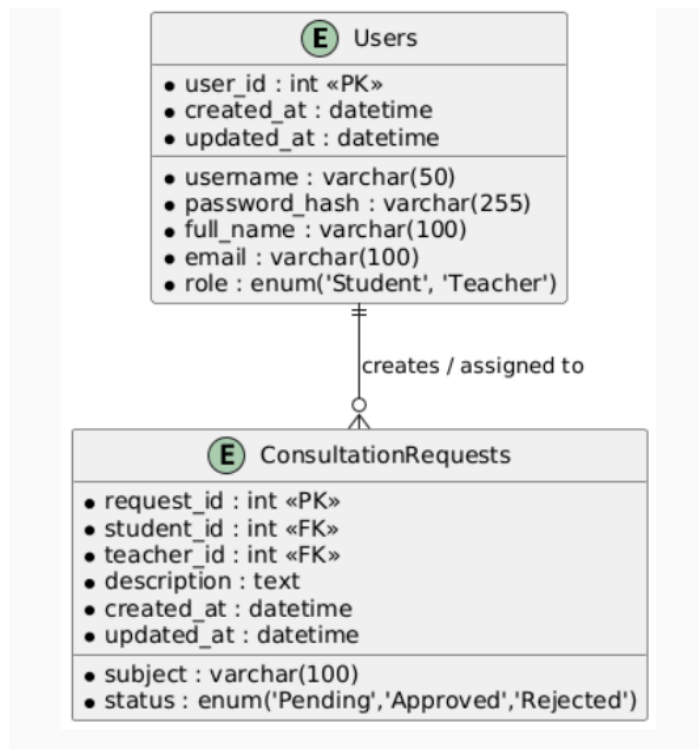
4. Non-Functional Requirements

- Security: Passwords hashed; protected pages require authentication.
- Performance: Handle multiple concurrent requests.
- Usability: Simple navigation for students and teachers.
- Reliability: Database integrity ensured for users and requests.

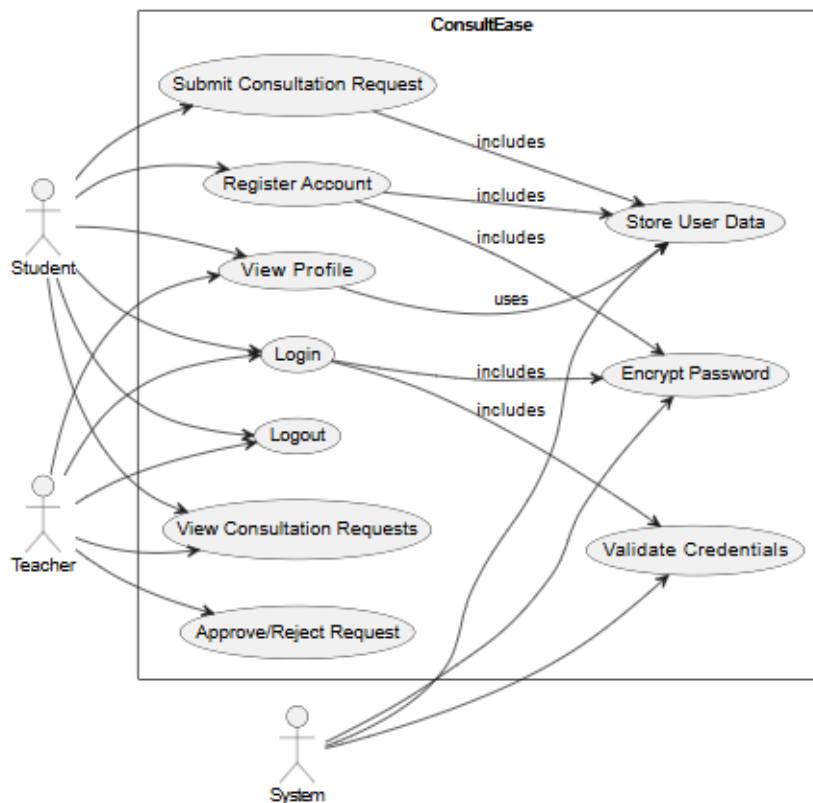
5. System Models (Diagrams)

Insert the necessary diagrams for the system:

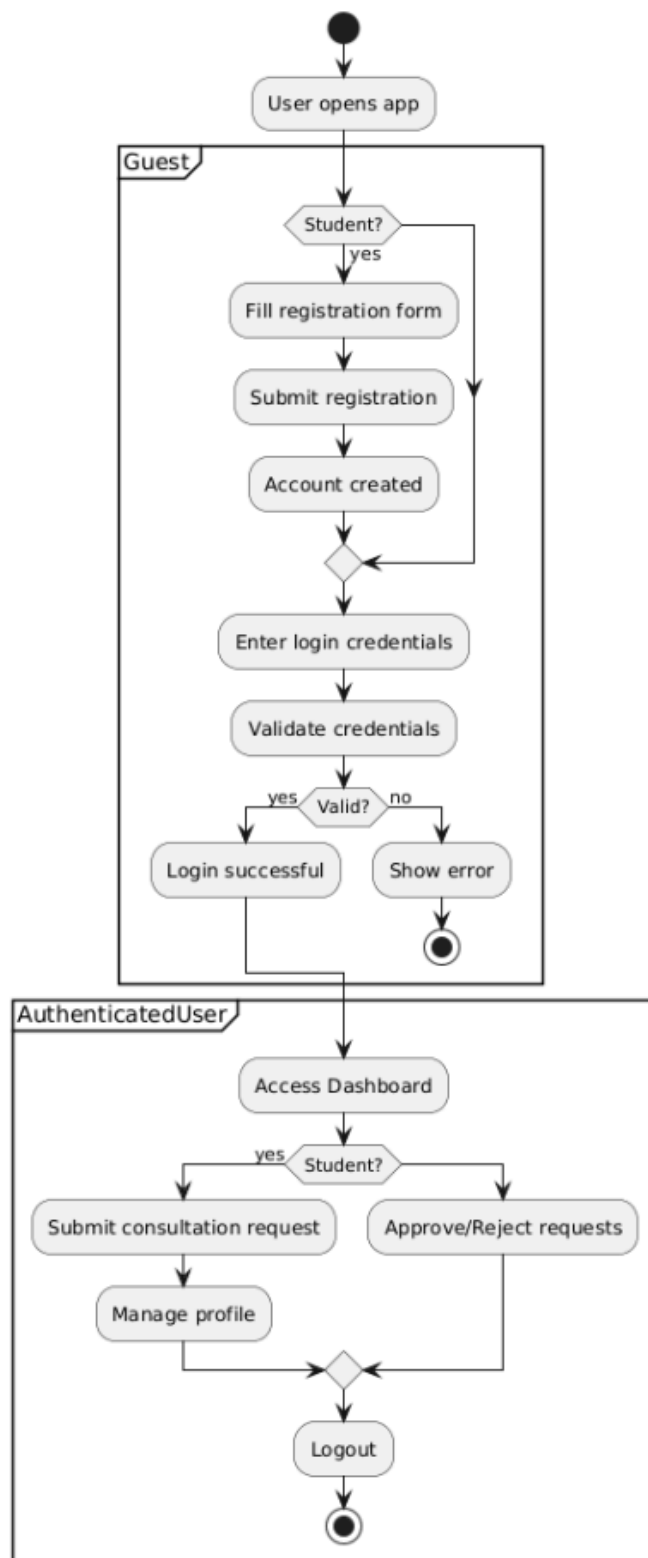
5.1. ERD



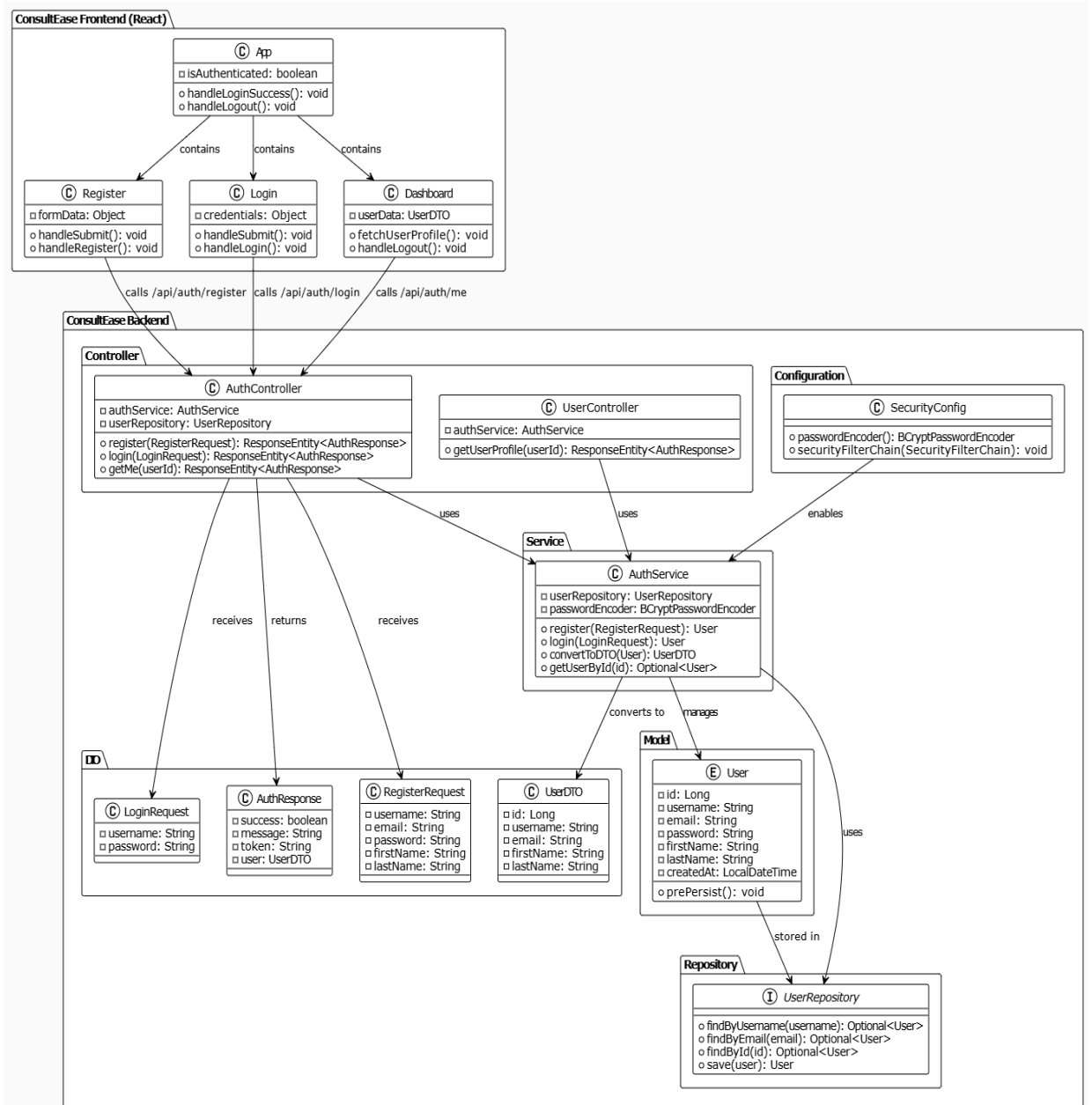
5.2. Use Case Diagram



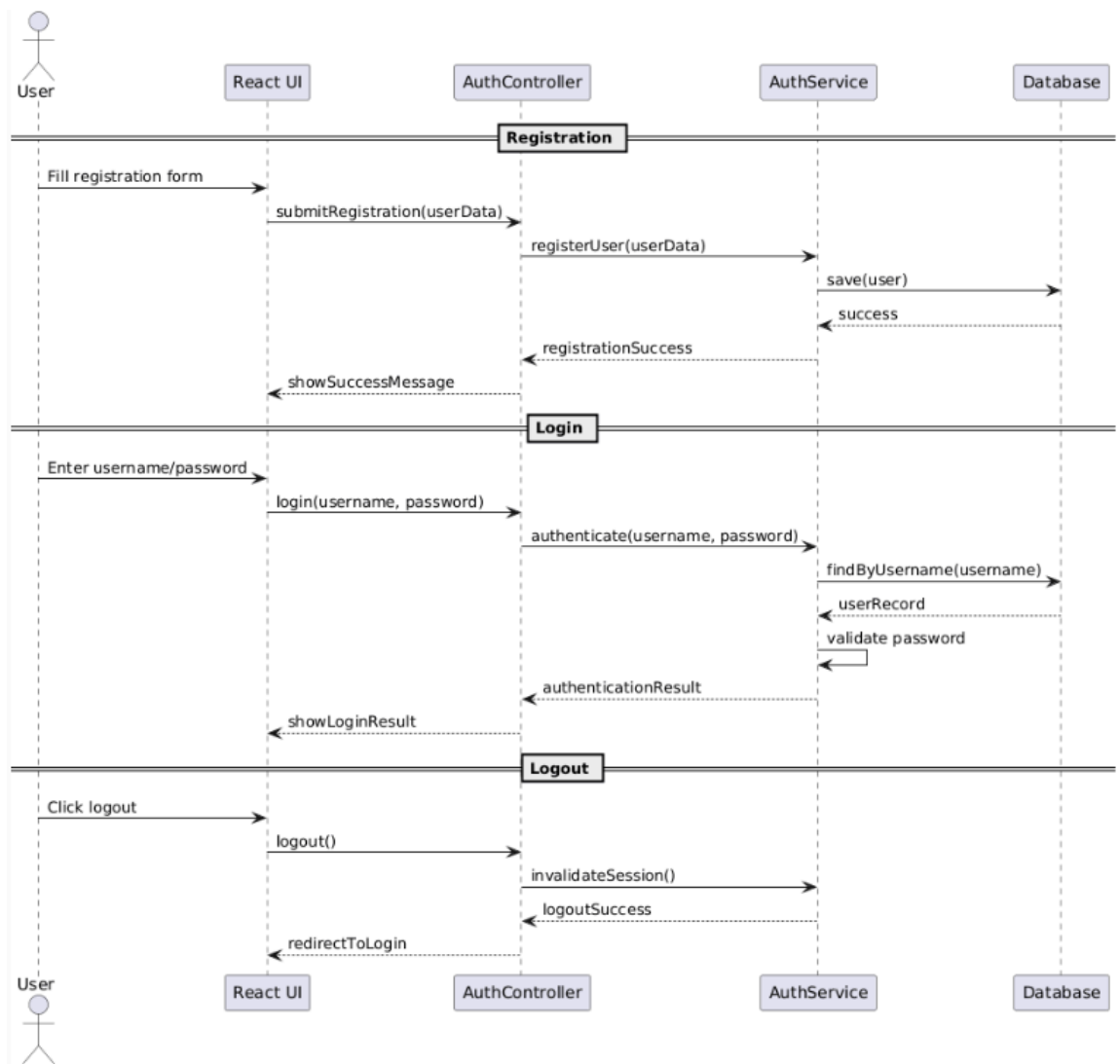
5.3. Activity Diagram



5.4. Class Diagram



5.5. Sequence Diagram



6. Appendices

Include any additional information, references, or support materials.